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// OSSP Skill 22 - Valgrind and Testing
// Demonstrates dynamic allocation, cleanup, and a small
// built-in test suite. Run this program with Valgrind to
// identify memory errors/leaks.
//
// Compile:
// gcc -Wall -Wextra -g 22_valgrind_test_demo.c -o skill22
//
// Run:
// valgrind --leak-check=full ./skill22

#include <stdio.h>
#include <stdlib.h>
#include <string.h>

static char *make_message(const char *name) {
    size_t size = strlen(name) + 32;
    char *msg = malloc(size);

    if (!msg)
        return NULL;

    snprintf(msg, size, "Hello, %s!", name);
    return msg;
}

static int test_message(void) {
    char *msg = make_message("OSSP");

    if (!msg)
        return 0;

    int ok = strcmp(msg, "Hello, OSSP!") == 0;
    free(msg);

    return ok;
}
```

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GNU nano 8.7.1

skill_22.c.c

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static int test_multiple_allocations(void) {
    char *a = make_message("Linux");
    char *b = make_message("Ubuntu");

    if (!a || !b) {
        free(a);
        free(b);
        return 0;
    }

    int ok = strstr(a, "Linux") && strstr(b, "Ubuntu");

    free(a);
    free(b);

    return ok;
}

int main(void) {
    printf("=== Skill 22: Valgrind + Testing ===\n");

    int passed = 0;
    int total = 2;

    if (test_message()) {
        printf("[PASS] test_message\n");
        passed++;
    } else {
        printf("[FAIL] test_message\n");
    }

    if (test_multiple_allocations()) {
        printf("[PASS] test_multiple_allocations\n");
        passed++;
    } else {
        printf("[FAIL] test_multiple_allocations\n");
    }
}
```

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    printf("\nResult: %d/%d tests passed.\n", passed, total);
    printf("Run with Valgrind to verify memory cleanup.\n");
```

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    return passed == total ? 0 : 1;
```

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}
```

```
sai_pavani@Sai-Pavani:~/OS_SKILL$ nano skill_22.c.c
sai_pavani@Sai-Pavani:~/OS_SKILL$ nano skill_22.c.c
sai_pavani@Sai-Pavani:~/OS_SKILL$ gcc skill_22.c.c -o skill_22
sai_pavani@Sai-Pavani:~/OS_SKILL$ ./skill_22
=== Skill 22: Valgrind + Testing ===
[PASS] test_message
[PASS] test_multiple_allocations

Result: 2/2 tests passed.
Run with Valgrind to verify memory cleanup.
sai_pavani@Sai-Pavani:~/OS_SKILL$ |
```